

NE 511: Multi-Physics of Nuclear Reactors

UNIT 3: Low Fidelity Multi-Physics Modeling

Lecture 16: SPACE-ENERGY DEPENDENT DYNAMICS: GENERAL DISCUSSION OF THE DYNAMICS PROBLEM; FLUX FACTORIZATION

WHY IS THE SPACE-ENERGY DEPENDENT TREATMENT NEEDED?

The point reactor results are not only inaccurate but also generally non-conservative.

The reasons are:

1. **Positive reactivity insertions**, which may cause or intensify an accident, are normally under-predicted by the neglect of the flux shape variation (i.e., by the point reactor model).
2. The value of a **negative counteracting reactivity**, on the other hand, is frequently over-predicted in the PRM.

SPACE-ENERGY DEPENDENCE IN FAST VS. THERMAL REACTORS

The flux distribution in a fast reactor is much less sensitive to spatial distortions than the flux distribution in a thermal reactor of the same power:

→ *Fast reactors have a lower built-in reactivity to compensate for fuel burnup and buildup of fission products.*

→ *Spatial flux distortions in fast reactor transients are generally smaller by about one order of magnitude.*

!! *The inclusion of the space dependence is more important for thermal reactor transient analysis.*

In thermal reactor transient analysis, the energy dependence of the neutron flux is approximated by a few-group model.

In a fast reactor, however, the energy dependence of the flux during accidents can vary considerably.

Since the Doppler coefficient is spectrum dependent, the time dependence of the fast reactor neutron spectrum must be treated in greater detail than in the few-group model.

General Discussion of the Dynamics Problem

DEPENDENCE OF THE MACROSCOPIC CROSS-SECTIONS ON THE FLUX

The basic neutronics problem of any transient analysis is the solution of the balance equation of the neutron flux and the delayed neutron precursor concentrations.

$$\frac{1}{v} \frac{\partial \Phi}{\partial t} = (F_p - M) \Phi + S_d + S$$

$$\frac{\partial C_k}{\partial t} = -\lambda_k C_k + F_k^d \Phi$$

+ transport and loss terms

$$S_d = \sum_k \lambda_k C_k \chi_k^d(E)$$

The microscopic cross-sections and atom densities depend on time, particularly through feedback effects.

Feedback is caused by changes in the neutron flux and the power → the operators are implicitly functions of the flux.

The dependence of the macroscopic cross-sections on the flux changes requires a solution of additional differential equations for:

- heat transfer,
- flow of coolant,
- cladding and fuel motion, etc.

The neutronics equations must be completed by models for the feedback phenomena.

MAJOR FEEDBACK PHENOMENA

The time dependence of the microscopic cross-sections results principally from the Doppler broadening of resonances with increasing temperature:

-- temperature dependence of "effective" cross-sections or group constants: $\delta\tilde{\sigma}_g = \delta\tilde{\sigma}_g[T(r, t)]$

The temperature field, $T(r, t)$, during a power transient is determined by the space distribution of the energy deposition and by the heat transfer.

The transfer of heat is strongly influenced by changes in the heat transfer media, e.g., by a reduction of coolant density or velocity:

$T(r, t)$ from $\left\{ \begin{array}{l} \text{power distribution} \\ \text{heat transfer} \\ \text{compositional changes} \end{array} \right.$

Thus, compositional changes must be treated in addition to or simultaneously with the variation of the temperature field.

MAJOR FEEDBACK PHENOMENA

The variation of the composition or the atom densities at a point r , $N_i(r, t)$, occurs either because of the variation of the local temperature, $T(r, t)$, which leads to expansion and possibly melting or boiling; or because of forced displacement.

The forced displacement is due to changes in the atom densities in spatial areas different from r , say r' , which then cause changes in the atom density at r .

$$\delta\Sigma[\delta N, \delta T] \begin{cases} \delta T(r, t) \\ \delta N_i(r', t) \end{cases} \left\{ \begin{array}{l} \text{loss of coolant} \\ \text{expansion} \\ \text{melting} \\ \text{boiling} \\ \text{cladding motion} \\ \text{fission gas ejection} \\ \text{fuel/coolant interaction} \\ \text{fuel slumping} \\ \text{fuel sweep out} \end{array} \right.$$

MAJOR FEEDBACK PHENOMENA

In many dynamics problems, the transport of precursors is not negligible as in the kinetics problems and in the simple dynamics problems treated above.

- The precursor density can change through motion and/or density change of fuel.
- If fuel elements are damaged or melt, precursor loss can occur through the loss of volatile components of fission products, particularly Bromine, Krypton, Iodine, and Cesium.
- Since different precursor groups contain different fractions of volatile precursors (e.g., precursor group 1 consists almost entirely of ^{87}Br), the precursor loss can vary strongly with delay groups.

GENERAL DISCUSSION OF THE DYNAMICS PROBLEM

- The equations that describe the neutronic and non-neutronic feedback effects represent the description of the complete nonlinear dynamics problem.
 - *Thus, the calculation of the neutron flux during a transient is only a part of the treatment of the complete dynamics problem.*
- Due to the strong coupling of the equations of the dynamics problem, the complete set of equations must be solved simultaneously.
- However, not all the phenomena listed above occur during the same phase of an accident → several phases of an accident progression can therefore be distinguished.

Example: Reactivity insertion (or coolant flow lost) with the entire scram system failure:
During the first phase, the time dependence of the neutron flux is influenced by the two feedback phenomena - Doppler effect and coolant temperature rise.

- **LWR** - the coolant density reduction has a negative reactivity effect; the Doppler effect provides a negative feedback.
- **Large LMFBR** - the coolant density reduction has mainly a positive reactivity effect, the Doppler effect provides a negative feedback.

GENERAL DISCUSSION OF THE DYNAMICS PROBLEM

In an accident without scram, the coolant may begin to boil with resultant voiding of fuel subassemblies:

- *After the onset of coolant boiling, cladding melting and cladding motion may occur.*
- *Fission gas may then be injected into the coolant and cause an increase in the voiding rate.*
- *Fuel melting might also occur during this phase.*
- *Molten fuel could be swept out of the core by the coolant.*
- *Fuel may also be driven out of the core by the pressure of fission gases enhanced by increased pressures due to fuel/coolant interaction.*
- *In some cases, the reactivity could be increased by fuel slumping (under gravity).*

All these phenomena have a strong effect on the neutron flux so that the compositional, temperature, and neutron flux variations must be treated simultaneously.

- In addition to the energy released during the accident, further energy production results from the decay heat of the fission products.
- The removal of the decay heat requires a post-accident heat removal.
- Possibility exists for the release of radioactivity.

All these events must be carefully evaluated in a complete safety analysis even though the probability of such accidents is very small.

Space-Energy Dependent Dynamics Approaches

Finite Difference Methods

Point Reactor Model

Modal Approach

Nodal Approach

FINITE DIFFERENCE METHODS

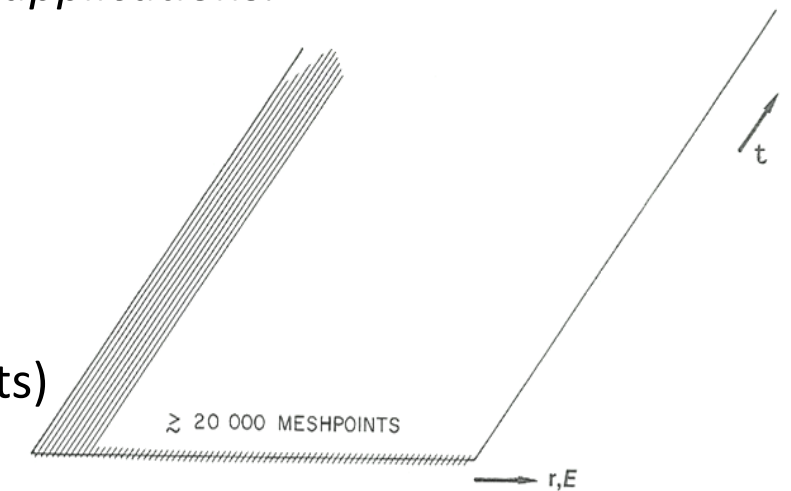
The differential and spatial differential operators in the balance equations are replaced by the corresponding finite difference quotients:

- The great advantage of the finite difference method is that for sufficiently small time steps it yields the correct solution of the time-dependent multigroup equations.
- The disadvantage is that the computation time is too large for normal practical applications.

Illustration of the finite difference procedure:

Each cross on the bottom line represents a flux value in the space-energy domain

A typical value of $\sim 20\,000$ space-energy points is needed for a description in two dimensions of the multi-group neutron flux (e.g., 20 groups $\times$ 40 $\times$ 25 mesh points)

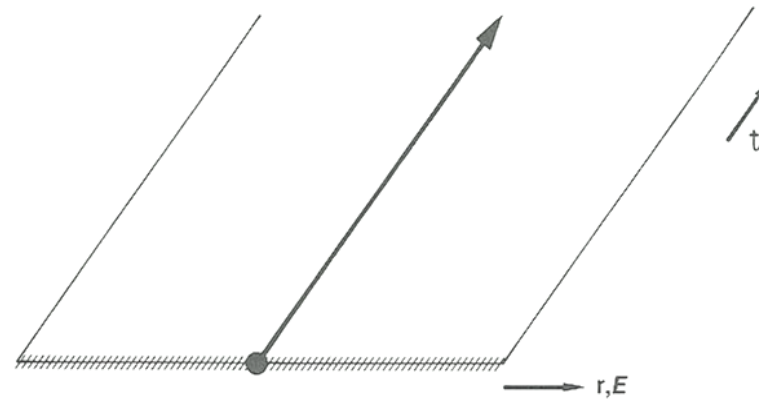


The time intervals may have to be very small to avoid numerical instability and obtain sufficient accuracy → thus, the number of flux values in the combined space-energy-time domain becomes very large, which leads to very long computation times.

In general, codes based on finite difference techniques either have no feedback capability or only feedback applicable to a limited range of thermal reactor dynamics problems.

POINT REACTOR MODEL

Only the time variation of the flux amplitude $P(t)$ is investigated under the influence of a time-dependent



Since the PRM generally yields non-conservative results, improvements or replacements are required for safety problem investigations.

MODAL APPROACH

The conceptual idea of the modal approach is to construct or synthesize the space-, energy (group)- and time-dependent neutron flux as a superposition of two or more components (modes):

- The desired savings in computation time is achieved by employing only a very few modes as well as by calculating the flux modes with simpler methods, for example, as solutions of static problems of normally lower dimensionality.
- The simplest form of a time-dependent flux synthesis is given by the expression:

$$\phi_g(r, t) = \sum_m h^{(m)}(t) f_g^{(m)}(r)$$

where the flux modes, $f_g^{(m)}(r)$, contain the space and group dependence, whereas the superposition coefficients, $h^{(m)}(t)$, are functions of time only.

- The use of only one term in the sum corresponds to the PRM if $f_g(r)$ is chosen as the initial multigroup flux shape.

The derivation of the equations for the $h^{(m)}(t)$ is, in principle, similar to the derivation of the point kinetics equations.

The flux expression of $\phi_g(r, t) = \sum_m h^{(m)}(t) f_g^{(m)}(r)$ is:

1. inserted into the time-dependent diffusion equation,
2. multiplied with weighting functions, and
3. integrated with respect to space and energy

In order to generate the appropriate set of independent equations, the number of the required weighting functions is equal to the number of unknowns, $h^{(m)}(t)$,

MODAL APPROACH

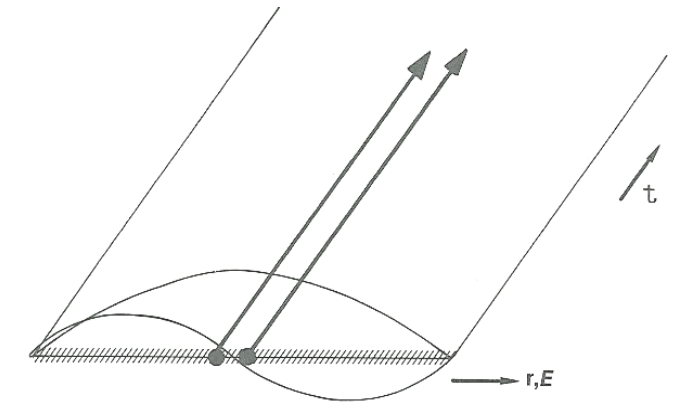
Illustration of the simplest form of the modal approach, two modes in a one-group approximation:

The all-positive curve describes the unperturbed flux.

The second mode is depicted by a sine function.

The superposition of the sine function with the unperturbed flux allows the perturbed flux to shift to the left or right, depending on the sign of the coefficient.

The two coefficients, $h^{(1)}(t)$ and $h^{(2)}(t)$, are to be determined as functions of time.



- The modal approach, applied to purely space-dependent dynamics problems, is often called the “space-time synthesis method”.
- Sometimes modal methods employing two or more energy groups are also called space-time synthesis methods (*indicates the dominance of spatial over spectral distortion of the flux in thermal reactor transients*).
- Synthesis methods have been developed for thermal reactor dynamics problems that may involve distorted spatial flux distributions.
- The description of the spectrum by two energy groups suffices in most cases.
- The synthesis approach has also been applied for fast reactor dynamics problems employing a large number of energy groups.

NODAL APPROACH

The conceptual idea of the nodal approach is to approximate the time-dependent flux by non-overlapping flux branches in a set of spatial sub-regions or "nodes":

$$\phi_g(r,t) \begin{cases} h^{(1)} f_g^{(1)}(r) & \text{in node 1} \\ h^{(2)} f_g^{(2)}(r) & \text{in node 2} \\ \vdots \end{cases}$$

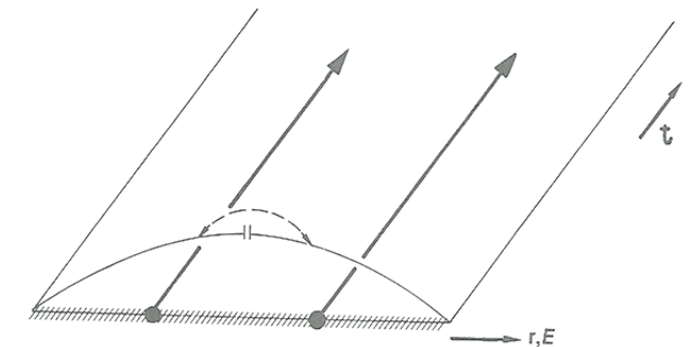
- If only one node is employed and if $f_g^{(1)}(r)$ is equal to the initial flux distribution, the nodal method reduces to the PRM.

Illustration of the nodal approach for a single energy group:

The spatial domain is subdivided into two nodes.

Neutron transfer between neighboring nodes leads to a coupling of the nodal flux branches.

If a flux tilt develops during the transient, i.e., if the flux branch in one node is shifted compared to the other branch, the nodal method then yields a somewhat discontinuous flux distribution.



- The development of the nodal approach is historically based on the kinetics of coupled reactors.
- If a system consists of two or more loosely coupled regions, the equations can describe the transient behavior fairly accurately.
- Two regions can be considered to be loosely coupled if the neutron transfer rate between them is very small compared to the neutron production rate in either region.

Flux Factorization Approaches

FLUX FACTORIZATION APPROACHES

The approaches that are formally closest to the familiar PRM are those based on flux factorization:

$$\phi_g(\mathbf{r}, E, t) = \underbrace{p(t)}_{\text{amplitude}} \underbrace{\psi(\mathbf{r}, E, t)}_{\text{time-dependent shape function}}$$

- Since the factorization allows for an arbitrary variation of the shape function with time, no approximation is introduced by representing the flux.
- Therefore, the same solution can be obtained from this approach as from the finite difference method.
- The shape function normally varies much more slowly with time and to a lesser degree than the amplitude function:
Flux factorization provides the opportunity to employ different intervals for the amplitude and shape function calculation.
- The computation time can be drastically reduced by choosing much larger time intervals between shape function calculations than between calculations of the amplitude function.
- Performing a shape function calculation only when needed does not practically reduce the accuracy of the solution.

FLUX FACTORIZATION APPROACHES

- There is only one type of time interval in the finite difference method, i.e., the one for the flux.
- This conceptually does not allow a benefit from the slow variation of the flux shape.
- The accuracy of the factorization approach is in part determined by the frequency of the shape function recalculation.
- If the shape function is calculated often in time, the factorization and the finite difference methods become equally accurate.

Let's recall that:

- (1) inserting the flux factorization into the time-dependent diffusion equation,
 - (2) integrating the resulting equation with respect to space and energy, and
 - (3) eliminating the explicit appearance of the shape function with a proper constraint condition
- results in the balance equation for the amplitude function alone:

$$\dot{p}(t) = \frac{\rho(t) - \beta(t)}{\Lambda(t)} p(t) + \frac{1}{\Lambda(t)} s_d(t)$$

FLUX FACTORIZATION APPROACHES

A second equation is required to determine the complete solution.

This second equation is obtained directly by inserting the flux factorization into the time-dependent diffusion equation:

$$(F_p - M)\phi(r, E, t) + S_d[\phi(r, E, t')] = \frac{1}{v} \frac{\partial}{\partial t} \phi(r, E, t)$$

The delayed neutron source results from the decay of precursors that were produced at earlier times, $t' \leq t$,

The space distribution of the delayed neutron precursors at time t can be expressed in terms of the flux at earlier times by formally solving the precursor concentration equation:

$$\frac{dC_k(r, t)}{dt} = -\lambda_k C_k(r, t) + \int_0^\infty \nu_{dk} \Sigma_f(r, E', t) \Phi(r, E', t) dE'$$



$$C_k(r, t) = C_k(r, 0) \exp(-\lambda_k t) + \int_0^t \exp[-\lambda_k (t - t')] \int_E \nu_{dk} \Sigma_f(r, E, t') p(t') \psi(r, E, t') dE dt'$$

FLUX FACTORIZATION APPROACHES

The delayed neutron source is then composed of the precursor concentration:

$$S_d(r, E, t) = \sum_k \chi_{dk}(E) \lambda_k C_k(r, t)$$

The dependence of the delayed neutron source on the previous flux is indicated by:

$$S_d(r, E, t) = S_d[p(t') \psi(r, E, t')]$$

Thus, the balance equation for the flux shape is given as:

$$[F_p - M] \psi(r, E, t) + \frac{1}{p(t)} S_d[p(t') \psi(r, E, t')] = \frac{1}{v} \frac{\dot{p}}{p} \psi(r, E, t) + \frac{1}{v} \frac{\partial}{\partial t} \psi(r, E, t)$$

We have a coupled system of equations for the two unknown functions, $p(t)$, and $\psi(r, E, t)$:

$$\dot{p}(t) = \frac{\rho(t) - \beta(t)}{\Lambda(t)} p(t) + \frac{1}{\Lambda(t)} s_d(t)$$

$$\begin{aligned} [F_p - M] \psi(r, E, t) + \frac{1}{p(t)} S_d[p(t') \psi(r, E, t')] = \\ = \frac{1}{v} \frac{\dot{p}}{p} \psi(r, E, t) + \frac{1}{v} \frac{\partial}{\partial t} \psi(r, E, t) \end{aligned}$$

FLUX FACTORIZATION APPROACHES

The amplitude function appears explicitly in two terms in the first equation, whereas the integral parameters and $s_d(t)$ in the second equation contain the shape function implicitly.

The equation which determines $p(t)$ depends only weakly and implicitly on ψ .

Alternately, the equation which determines ψ depends only weakly on $p(t)$.

This pair of coupled equations is solved simultaneously with the solution at all times subjected to the constraint condition:

$$\int_V \int_0^\infty \frac{\phi_0^*(\mathbf{r}, E) \psi(\mathbf{r}, E, t)}{v(E)} dV dE = K_0$$

Since the coupled equations depend nonlinearly on $p(t)$ and ψ , the basic solution method is similar to that often applied in solving nonlinear systems of differential equations of this type.

One function is assumed known; this results in a linear equation for the other function, which is solved first.

The latter function is then assumed known, and the resulting linear equation for the former is solved.

FLUX FACTORIZATION APPROACHES

These three equations, the pair of coupled equations and the constrain condition, are equivalent to the original time-dependent diffusion equation since no approximation has yet been introduced.

A sequence of approximations is introduced below leading to a sequence of methods, which ranges from the complete numerical solution to the PRM.

The first simplification ("approximation") consists of introducing different time steps for the amplitude and shape functions.

Since the shape function varies slowly with time, its time derivative may be calculated by a backward difference:

$$\frac{\partial \psi(r, E, t)}{\partial t} \approx \frac{1}{\Delta t_s^s} [\psi(r, E, t_s^s) - \psi(r, E, t_{s-1}^s)]$$

where the time interval $\Delta t_s^s = t_s^s - t_{s-1}^s$ is generally very much larger than the corresponding time step required to solve the amplitude equation.

FLUX FACTORIZATION APPROACHES

Inserting the last equation into the balance equation for the flux shape and collecting the terms containing only previous times on the right side yields the equation for the shape function in the so-called [improved quasi-static method](#):

$$\left(F_p - M - \frac{1}{\nu \Delta t_s^s} - \frac{1}{\nu} \frac{\dot{p}}{p} \right) \psi(r, E, t^s) = -\frac{1}{p} S_d [p(t') \psi(r, E, t')] + \frac{\psi(r, E, t_{s-1}^s)}{\nu \Delta t_{s-1}^s}$$

The development of the improved quasi-static method was preceded by the [quasi-static method](#) in which the shape function derivative in the balance equation for the flux shape is neglected.

This approximation leads to the following balance equation for the shape function in the quasi-static method:

$$\left(F_p - M \frac{1}{\nu} \frac{\dot{p}}{p} \right) \psi(r, E, t^s) = -\frac{1}{p} S_d [p(t') \psi(r, E, t')]$$

FLUX FACTORIZATION APPROACHES

After the neglect of the derivative $\frac{\partial \psi}{\partial t}$, the time variable t appears only as a parameter in the balance equation for the flux shape.

Thus, the problem is formally a static problem with cross sections depending parametrically on time.

The left side of the last equation is augmented by an α/v term ($\alpha = p/\dot{p}$ is the inverse period), which represents the residue of the time derivatives in the balance equation for the flux shape.

It treats the problem as if the flux shape were static.

The delayed neutron source, however, depends on the entire flux history and thus on earlier times, $t' \leq t$.

If the delayed neutron source is approximated at time t in a quasi-stationary fashion as a fraction of the prompt source, and if the α/v term is neglected, the specific kinetic features in the shape equation are completely eliminated.

An eigenvalue, λ , must be incorporated to obtain a nontrivial solution; this results in the balance equation for the adiabatic method: $(\mathbf{M} - \lambda \mathbf{F})\psi = 0$

Neglecting also the time dependence of the operators $\mathbf{M}$ and $\mathbf{F}$ in calculating the shape function results in the balance equation for the point reactor model: $(\mathbf{M}_0 - \lambda \mathbf{F}_0)\psi_0 = 0$

FLUX FACTORIZATION APPROACHES

The sequence of factorization methods is listed below - the first column describes certain features, and the body of the table indicates whether or not these features are accounted for in the various methods.

	Model		
	Direct Numerical	Improved Quasi-static	Quasi-static
$\partial \psi / \partial t$	Yes	Yes	No
α / ν	Yes	Yes	Yes
$S_d[p(t')\psi(r, E, t')]$	Yes	Yes	Yes
Implicit coupling via feedback	Yes	Yes	Yes
Decoupled shape distortion	Yes	Yes	Yes
Initial shape	Yes	Yes	Yes
	Model		
	Adiabatic	Adiabatic with pre-calculated shape function	PRM
$\partial \psi / \partial t$	No	No	No
α / ν	No	No	No
$S_d[p(t')\psi(r, E, t')]$	No	No	No
Implicit coupling via feedback	Yes	No	No
Decoupled shape distortion	Yes	Yes	No
Initial shape	Yes	Yes	Yes

FLUX FACTORIZATION APPROACHES

- The first two methods (direct numerical and improved quasi-static) are mathematically equivalent in the sense that no approximations except finite differencing are introduced.
- Thus, both methods yield the same result for sufficiently small time steps.
- The accuracy of the other methods generally decreases with decreasing sophistication of the shape function balance equation.
- Both adiabatic methods listed in the table calculate the flux shape from the same type of balance equation, $(\mathbf{M} - \lambda \mathbf{F})\psi = 0$.
- The simpler of the two methods allows only changes in the reactor for which the sequence of configurations is known in advance.

FLUX FACTORIZATION APPROACHES

- Examples are the motion of control rods or fuel subassemblies.
- For such known sequences of motion, the flux distortion and its effect on the reactivity can be pre-calculated.
- The resulting time-dependent reactivity is inserted into the point kinetics equation.
- This method represents an obvious improvement over the PRM.
- If there are no other changes in the reactor except the pre-calculated configurations, the results may be fairly accurate.
- If, however, feedback plays an important role, the reactor may exhibit a number of changes - the combination of which cannot be known in advance, that is, without actually solving the initial value problem.

Next Class: *Space-Energy Dependent Dynamics:*

Quasi-Static Methods

Kinetics vs. Dynamics

Dynamic Reactivity Coefficients